Reg. No. | | | | | | | | | | |

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER B.TECH. (INSTRUMENTATION AND CONTROL ENGG.)

END SEMESTER DEGREE EXAMINATIONS, NOVEMBER - 2019

SUBJECT: BIOMEDICAL INSTRUMENTATION [ICE 3102]

TIME: 3 HOURS MAX. MARKS: 50

Instructions to candidates :*Answer ALL questions and missing data may be suitably assumed.*

1A. What are the advantages of using needle electrodes when compared to surface electrodes? List the types of surface electrodes.

1B. Draw and explain functions of different blocks of Electrocardiograph.

1C. How we can get truncated waveform in a cardiac defibrillator? Explain with a circuit diagram.

1D Explain working of competitive pacemaker with a block diagram.

(2+4+2+2)

2A Explain the procedure to remove kidney stone using ultrasound with suitable diagram.

2B Discuss in detail 10-20 electrode system used for recording of EEG.

2C Draw a typical ECG waveform and mark the different waves and time intervals. What do the different waves indicate? What is the significance of R-R interval in an ECG signal?

(3+4+3)

3A Derive an expression relating blood flow rate and induced emf with a suitable diagram.

3B What are the characteristics and uses of X-rays? With a block diagram explain different components of CT scanner.

3C Write short notes on ventillators and respirators.

(3+4+3)

4A Explain the following terms with respect to NMR imaging

i) Free Induction Decay ii) Excitation iii) Emission

4B Explain with the help of a block diagram, working of MRI instrumentation system. List two application of MRI imaging system

4C With a block diagram explain different components of X-ray tube.

(3+4+3)

5A What is Incubator? Explain different types of Incubators.

5B Explain a non-invasive technique used to monitor the serum bilirubin level of a newborn baby.

5C Write short notes on the following

i) Apnea Monitor

ii) Endoscopes

(3+3+4)

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